Sentiment Analysis and Sarcasm Detection in Product Reviews using NLP and LLMs

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Aims & Objectives

AIM:
To analyse and compare different models for sentiment analysis and sarcasm detection, and to understand how sarcasm affects classification. This study also aims to improve how customer reviews are interpreted in real-world application such as businesses, recommendation system, and research.

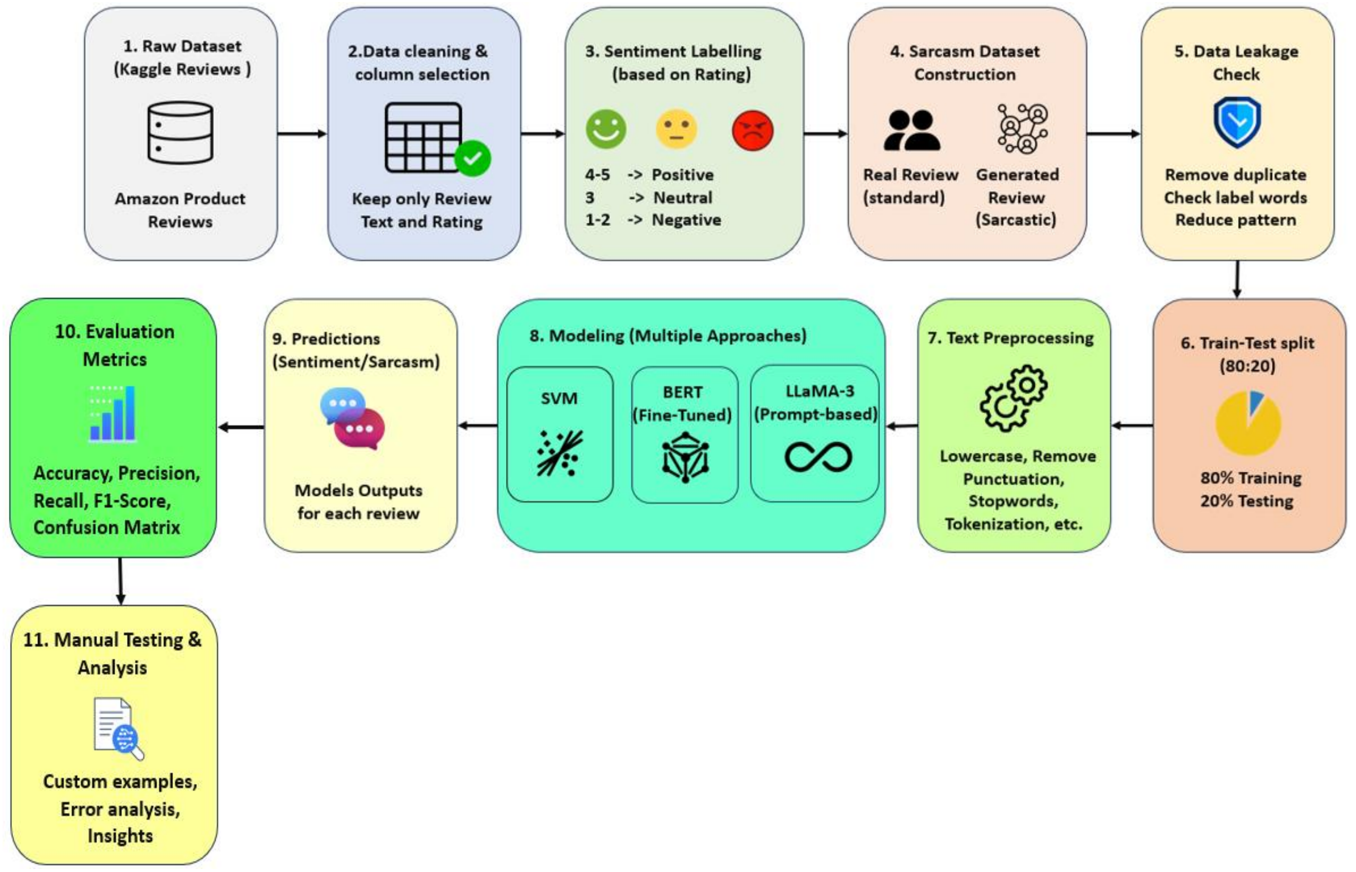
OBJECTIVES:
- To study how sarcasm affects sentiments in reviews.
- To compare different NLP models in understanding user intent.
- To evaluate whether advanced models can capture contextual meaning beyond keywords.
- To explore how these techniques can improve the reliability of automated review analysis system

Methods

Methodology:
- Dataset collection from Kaggle.
- Data cleaning and preprocessing
- Sentiment labels created from rating
- Sarcasm dataset prepared
- Models trained and evaluated

Models Used:
- SVM (baseline models)
- Fine-tuned BERT
- Zero-shot Bert
- LLaMA-3

Dataset Type	Size	Labels/Classes
Product reviews	~1000	Positive, Negative, Neutral
Sarcastic and non-sarcastic reviews	~1000	Sarcastic, non-sarcastic

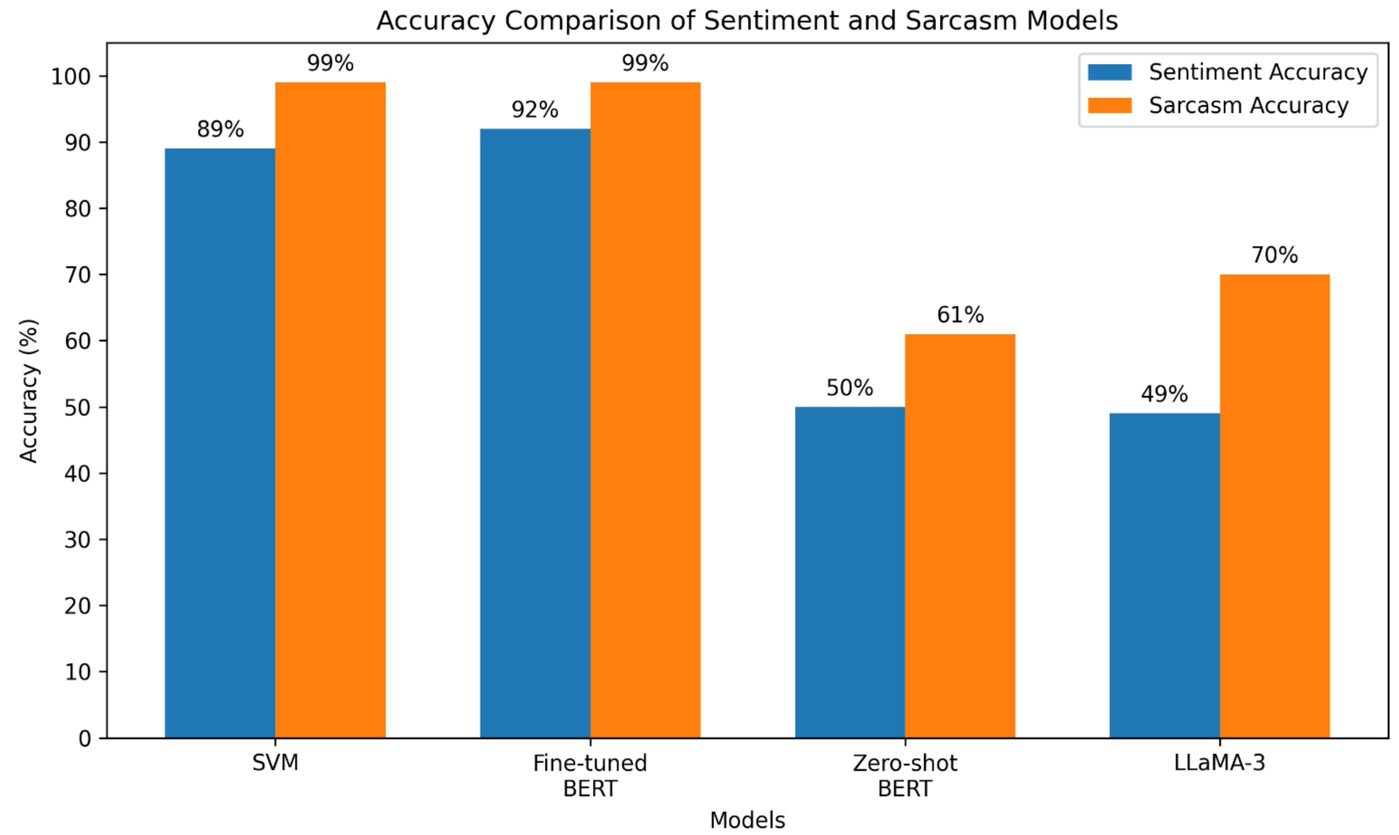


Ongoing Work

- Improving model performance on subtle and complex sarcastic reviews.
- Refining preprocessing techniques to reduce noise in the data.
- Testing models on additional unseen examples.
- Analysing model predictions to better understand errors.
- Exploring better prompt design for LLMs to improve consistency in prediction.
- Comparing model behaviours across different types of reviews, including short and long text.

Results

Results:
- SVM: 89% (sentiment), 99% (sarcasm)
- BERT: 92% (sentiment), 99% (sarcasm)
- Zero-shot BERT: 50% (sentiment), 61% (sarcasm)
- LLaMA-3: 49%(sentiment), 70% (sarcasm)



Predicted Outcomes & Conclusion

Prediction: It was expected that advance models like BERT and LLaMA-3 would perform better than traditional models due to its ability to understand context.

Outcome:
- The results showed that BERT achieved the highest accuracy, while SVM relied on word pattern
- Zero-shot BERT and LLaMA-3 showed better generalisation but lower accuracy.

Overall sarcasm remains a challenging problem in sentiment classification and reduces accuracy. Bert performs best overall, while LLMs show better generalisation. A combined approach can improve real-world application.

Acknowledgements

I would like to express my sincere gratitude to my supervisor for their guidance, support, and valuable feedback throughout this project. I would also like to thank my university for providing the resources and learning environment that made this work possible.

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